

Using video modeling to teach play comments to dyads with ASD

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Individuals with autism spectrum disorder (ASD) often display deficits in social and conversational skills. One method used to improve social deficits includes video modeling. When targeting conversational skills, few studies have included individuals with ASD as conversational partners. We evaluated the effects of video modeling on commenting (i.e., scripted statements) during leisure activities with dyads of children with ASD using a multiple-probe-across-activities design. Video modeling alone was found to be effective in increasing scripted statements for 3 of 6 participants. Video modeling, tangible reinforcement, and additional prompts were necessary for the remaining participants. Results maintained 1 and 3 weeks after mastery. Procedures, goals and outcomes were reported as socially valid. Limitations and areas of future research are discussed.

Key words: autism spectrum disorder, commenting, leisure activities, video modeling

The characteristics of autism spectrum disorder (ASD) include language and social skill deficits that are barriers to developing meaningful and social relationships (American Psychiatric Association, 2013; Murdock, Ganz, & Crittendon, 2013). For example, deficits in spontaneous and reciprocal language can be a barrier for social interactions during school, family dinners, or community outings (Green et al., 2013; McGee, Almeida, Sulzer-Azroff, & Feldman, 1992). Given these deficits, it is important to develop methods to teach social skills in the context of activities that typically include social interactions.

Video modeling has been shown to be effective in teaching a variety of social skills to children diagnosed with ASD during social

activities (Shukla-Mehta, Miller, & Callahan, 2010). One common strategy is to teach social skills to children in the context of play, which might be beneficial because it may capitalize on motivation and natural communities of reinforcement. Research has shown improvements in reciprocal pretend-play (MacDonald, Clark, Garrigan & Vangala, 2005), conversational speech (Charlop & Milstein, 1989), giving compliments (Apple, Billingsley, & Schwartz, 2005), and perspective taking (LeBlanc et al., 2003) using video modeling in play contexts.

Although there is evidence supporting the use of video modeling to teach social skills, there are limitations to the existing research. First, several studies have incorporated additional instructional procedures besides video modeling (e.g., Carlile, DeBar, Reeve, Reeve, & Meyer, 2018; Kourassanis, Jones, & Fienup, 2015), and few studies have evaluated video modeling as a sole intervention (e.g., MacDonald et al., 2005; Nikopoulos & Keenan, 2007). Second, little is known about how to program for the generalization of trained skills to untrained contexts. For example, Jones, Lerman, and Lechago (2014) found that social responses

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trained with adults did not occur when children interacted with peers who had not participated in training. This finding raises concerns about the generality of behavior change produced by video modeling with adults (Jones et al., 2014; Nikopoulous & Keenan, 2003, 2007), siblings (Reagon, Higbee, & Endicott, 2006; Taylor, Levin, & Jasper, 1999), or peers of typical development (MacDonald, Sacramone, Mansfield, Wiltz, & Ahearn, 2009) as conversation partners during training. Since children with ASD may have more opportunities to interact with peers with ASD than with adults or peers of typical development, it may be important to include children with ASD as conversational partners when teaching social skills. However, only a few studies have included individuals with ASD as conversational partners (Betz, Higbee, & Reagon, 2008; Charlop & Milstein,1989; Kourassanis et al., 2015; McGee, Krantz, & McClannahan, 1984), and of those studies, few have assessed the sustainability of improvements in social skills over time (e.g., MacDonald et al., 2009; Nikopoulos & Keenan, 2007) or across untrained peers or activities (e.g., Jones et al., 2014; Reagon et al., 2006).

The purpose of this study was to extend previous research by evaluating the effectiveness of video modeling on increasing scripted

statements and on-task behavior during leisure activities with dyads of individuals diagnosed with ASD. We also assessed whether participants made scripted statements when playing with a peer who was not part of training and whether they continued to make statements during play 1 and 3 weeks after mastery. Finally, social validity of the procedures and outcomes were assessed.

METHOD

Participants and Setting

Three dyads of children diagnosed with ASD participated. Dyads consisted of peers who were similar in age, attended the same school, and demonstrated mastery of similar leisure activities. Dyads included Leila and Adam, Terry and Sean, and Henry and Joshua. All children were between 5 and 9 years old and received ASD diagnoses from an outside agency. See Table 1 for participant assessment outcomes across language assessments (i.e., Peabody Picture Vocabulary Test-Fourth Edition; Dunn & Dunn, 2007; the Expressive Vocabulary Test-Second Edition; Williams, 2007), an autism rating scale (i.e., the Gilliam Autism Rating Scale-Third Edition; Gilliam, 2013), and a criterion-referenced skill assessment (i.e., Verbal

Table 1
Participant Assessment Outcomes across Dyads

Participant	Age	PPVT	EVT	GARS	VB-MAPP		
		Standard Score	Standard Score	Autism Index Score	Raw ESSA	Raw Barriers	Raw Mile-stones
Dyad 1							
Leila	9:4	94	78	65	97	12	168
Adam	9:2	60	75	48	100	7	162
Dyad 2							
Terry	7:4	63	64	94	100	18	148.5
Sean	7:9	59	61	92	96	32	117
Dyad 3							
Henry	5:7	60	73	87	98	13	137
Joshua	7:7	73	67	90	99	27	122.5

Note. PPVT = Peabody Picture Vocabulary Test - Fourth Edition (Dunn & Dunn, 2007; EVT = Expressive Vocabulary Test- Second Edition (Williams, 2007); GARS = Gilliam Autism Rating Scale - Third Edition (Gilliam, 2013); VB-MAPP = Verbal Behavior Milestones Assessment and Placement Program (Sundberg, 2008).

Behavior Milestones Assessment and Placement Program; Sundberg, 2008) across dyads.

Inclusionary criteria required (a) a diagnosis of ASD, (b) vocal behavior of at least one- to two-word utterances, (c) a generalized imitative repertoire based on report and direct assessment, (d) an individualized education program goal related to the acquisition of conversational skills, and (e) at least five mastered leisure activities (e.g., the game Zingo™) verified by the first author. For the purpose of this study, all leisure activities were limited to previously mastered games. Participants attended a private school which used intervention strategies based on the principles of applied behavior analysis. Sessions occurred in a conference room at the private school.

Dependent Variables and Interobserver Agreement

The primary dependent variable was the percentage of scripted statements emitted. Scripted statements were defined as statements that matched each participant's respective script. Similar to Krantz and McClannahan (1993), a statement was scored as correct if it matched the script except for an omission of a conjunction, pronoun, preposition, or an article or a change of verb tense. For example, if the script stated, "I'm still going to win," "I'm going to win" was scored as a correct scripted statement. If a scripted statement was stated more than once, repetitions were not scored. Participants sat so their positions corresponded with their respective models. For example, the participant who sat on the right side followed the model positioned on the right. Observers scored scripted statements as correct when the participant vocalized the statement. Data were collected on the correct occurrence of the scripted statement and were calculated by dividing the number of correct scripted statements by the total number of statements (i.e., 10) and this quotient was converted to a percentage. Data were summarized as the percentage of correct scripted statements.

Prior to the study, experimenters observed age-matched peers of typical development to generate scripts. During observations, these peers played the target games (i.e., Candyland™, Zingo™, Memory™, and Fruit Ninja™) for a maximum of 10 min. The first author transcribed the contextual vocal and motor behaviors emitted by peers while playing and generated scripts for each game (see Table 2 for a representative script; others are available upon request from corresponding author). Statements emitted by the typically developing children were categorized across initiations to play, assertive statements, compliments, and statements of play termination. Scripts were then created to include these components.

A secondary dependent variable included the percentage of intervals of on-task behavior. Observers scored on-task behavior when the participant vocalized contextually appropriate statements (e.g., saying "Your turn" even if it was not part of the script) including scripted statements, oriented to the game or partner, and/or appropriately manipulated materials related to the activity (e.g., moving the appropriate pieces). Nonexamples included moving more than 0.30 m away from the designated play area, which was assessed using an

Table 2
Scripted Statements across Participant Used during
Candyland™

Leila	Adam
"Come play Candyland™ with me"	"Umm okay"
"I'm being blue"	"I'll be red"
"I got red"	"Hooray! I got green"
"Awesome job"	"I got the blue lollipop"
"I'm going to pass you"	"Nice move"
"Awesome I got the ice cream!"	"You're good at this game"
"If you get the peanut you have to back to the start"	"I got yellow"
"I'm going to win"	"I hope I win!"
"I had fun!"	"Me too"
"Bye"	"Bye"

additional conference table as a measure, and not responding within 2 s of a peer’s initiation. Observers recorded on-task behavior using 10-s momentary time sampling with a 3-s observation window. Data were summarized as a percentage of intervals with on-task behavior by dividing the numbers of intervals with on-task behavior by the total number of intervals and converting this quotient to a percentage.

A second data collector independently scored the dependent measures from video recordings for a total of 51% (Leila & Adam), 50% (Sean & Terry), and 51% (Henry & Joshua) of sessions across all phases. Interobserver agreement (IOA) was collected on each occurrence of a scripted statement. For scripted statements and on-task behavior, an agreement was scored when both the primary and secondary observer recorded the occurrence (+) or nonoccurrence (-) of target responses. Observers calculated IOA data by dividing the number of agreements by the number of agreements plus disagreements and converting to a percentage. Mean IOA was at least 80% (range, 80–100%) across all conditions and dyads for scripted statements and on-task behavior.

Preassessments

Vocal and motor imitation assessment. A vocal and motor imitation assessment was conducted with the dyads across two different activities (i.e., Tic Tac Toe™, and dressing up and singing) to assess the participant’s generalized imitative repertoire when a video model was presented on an iPad™. Each video model was 1 min long and showed two adults modeling a sequence of 10 motor responses and scripted statements not targeted in the study (see Table 3 for dress up targets; a list of the responses targeted during Tic Tac Toe™ are available upon request). Video models were presented twice (MacDonald et al., 2009). Observers recorded correct imitations of the motor responses or scripted statements modeled

Table 3
Tic Tac Toe Motor and Vocal Components

Leila		Adam	
Statements	Motor components	Statements	Motor components
“I’m going to be orange”	Picks up orange marker	“I’ll be red”	Picks up red marker
“I’m going to go first”	Looking at peer	“Okay”	Looking at peer
“I’m going to put my x in the middle”	Places X in the middle	“I’ll put my O here”	Draws O above
“That’s the best spot”	Faces peer	“Next best spot is here”	Draws O in bottom left corner
“Now I can fill the whole board”	Faces peer	“And I go right here”	Draws O in middle right column
“I’ll put my x right here”	Places X above O	“I’ll go under here”	Draws O in bottom row, middle column
“I’m going to win”	Faces peer	“I’ll go up here”	Draws O in top right corner
“And I’ll put it here”	Places X in middle top row	“Darnit!”	Facing peer
“I’ll go right here!”	Places X in bottom right corner	“You won!”	Places in bottom right corner
“Told ya!”	Draws line across	“See ya!”	Faces peer

in the video. When participants imitated at least 80% of target responses across two consecutive sessions, the skill was considered established.

Two (Terry and Joshua) of the six participants required preteaching to improve imitation skills. During the initial assessment, the mean percentage of correct imitations was 50% for Terry across both games; Joshua imitated 60% (dress up) and 50% (Tic Tac Toe™) of models on average. To improve imitation skills, the experimenter prompted motor responses using graduated guidance (Torres, DeBar, Reeve, Meyer, & Covington, 2018) and presented scripted statements on a Mini-me audio device™. Tokens were delivered for correct imitative responding on a fixed-ratio (FR) 1 schedule using participants’ individualized

reinforcement system. Following prompting and reinforcement, Terry's imitation improved and mastery was achieved within four (dress up) sessions and responding generalized to Tic Tac Toe™. Joshua mastered dress up within two sessions and generalization of vocal and motor responses to Tic Tac Toe™ occurred. Leila, Adam, Sean, and Henry imitated 100% of the vocal and motor components across both dress up and Tic Tac Toe™.

Preference assessment. The Reinforcer Assessment for Individuals with Severe Disabilities (Fisher, Piazza, Bowman, & Amari, 1996) was presented to instructors to identify preferred activities and items to be included in a brief multiple-stimulus without replacement (BMSWO; Carr, Nicolson, & Higbee, 2000) preference assessment. Experimenters conducted these assessments to identify highly preferred items that were provided to the participants for session participation. At the end of each session, participants selected one item from an array of pictures of preferred items. Highly preferred items included an iPad™, clay, and popcorn for Leila, and a Nintendo 3DS™, an iPad™, and Legos™ for Adam. For Terry, highly preferred items included an iPad™, magnetic sticker book, and a book, while Sean's highly preferred items included iPad™, Oreos™, and the game Operation. The BMSWO identified M&M's™, an iPad™, and drawing for Henry, and an iPad™, apples, and a puzzle as highly preferred for Joshua.

A single-stimulus engagement preference assessment (Hagopian, Rush, Lewin, & Long, 2001) was conducted with mastered leisure activities to assess engagement with activities. Total duration of engagement was collected by recording the duration of appropriate manipulation of materials (i.e., manipulation in the manner in which the activity materials were intended). The observers recorded appropriate manipulation of materials in the absence of stereotypy and/or disruptive behavior (e.g., crying,

screaming, whining) with an onset and offset of 5 s while remaining in the designated area (within 0.30 m). Sessions were terminated when one of the participants was no longer engaged for at least 5 s (e.g., inappropriately manipulating the materials or engaging in stereotypy) or after 10 min of engagement. Leisure activities with at least 3 min of engagement were included in experimental sessions. Games selected for Leila and Adam were Fruit Ninja™, Candyland™, and Memory™. For Sean, Terry, Henry, and Joshua the games selected were Zingo™, Memory™, and Fruit Ninja™.

Experimental Design and General Procedures

A multiple probe design across leisure activities was used to evaluate the effects of video modeling on scripted vocalizations and on-task behavior (Horner & Baer, 1978). During baseline, intervention, generalization, and maintenance, participants were seated so their position corresponded with the models in the video. As with MacDonald et al. (2009), reinforcement was not provided contingent on correct responding (i.e., imitation of the scripted statements), unless introduced as a procedural modification. An individualized reinforcement system, similar to the one used in their educational setting, was created and used to reinforce attending behavior (i.e., appropriate orienting to the video, materials, or peer) on a variable-interval 1-min schedule. For example, Layla's reinforcement system consisted of earning and exchanging 10 tokens prior to accessing choice of a preferred item. Sessions were conducted one to three times per day and 3 to 5 days per week. Sessions were a maximum of 5 min unless terminated early due to problem behavior. Three sessions were terminated for Terry and Sean across baseline and video modeling because of aggressive behavior. Sessions began after the experimenter said, "Play the game," and ended when the participants finished

playing the game or when a session was terminated.

Baseline. At the beginning of the session, dyads were presented with target materials and given the direction to “Go play.” No video modeling or reinforcement was provided.

Video modeling. Four video models from a third-person perspective were created and included two adults familiar to the participants (i.e., instructors at the school). Video models showed the instructors playing appropriately (e.g., moving materials in the manner they were intended) and engaging in a reciprocal conversation across 10 conversational exchanges. Reciprocal conversations included a social initiation, compliment, assertive statement, and statement of activity termination. Videos were 1 min 20 s (Candyland™), 1 min 38 s (Zingo™), 1 min 35 s (Memory™), and 1 min 15 s (Fruit Ninja™). Videos were presented to each participant via an iPad™. A Google Fire Tablet™ was used to video record sessions. An iPad™ was presented to participants along with the materials specific to the leisure activity. Participants were instructed to “Watch the video,” and were then given the direction to “Go play” after watching the video one time. Once the participants performed at a criterion level with the presentation of the video for two consecutive probes, the video was removed for subsequent sessions. Scripted statements were considered mastered once participants emitted 90% of target responses across two consecutive sessions.

Procedural modifications. Based upon visual analysis, when data for scripted statements were stagnant (i.e., were not improving or were progressing in an incompatible direction) during video-modeling, the experimenter delivered highly preferred edibles in a cup on an FR 1 schedule for correct scripted statements. Prior to the start of sessions using edibles, experimenters provided a choice of highly preferred edibles identified from the BMSWO. Edibles were introduced during the 10th session of

Zingo™ for Henry and Joshua, during the 11th intervention session of Memory™ and the 67th session of Fruit Ninja™ for Joshua. For Sean, edibles were also implemented during the 16th session of Memory™ and the 47th session of Zingo™. When edibles were not effective in producing criterion responding, an in-vivo model (i.e., vocal prompt) was introduced. Vocal prompts were faded across (a) full vocal prompts (e.g., “Say this is awesome” or “Bam”), (b) partial vocal prompts (e.g., “This is” or “Ba”), and phonemes (e.g., “Th” or “B”). At least 90% of trials with correct responses for two consecutive probes were required before the next level of prompting was introduced. When these criteria were met with phoneme prompts, vocal prompts were no longer presented. When the participants emitted the statements in the absence of the vocal prompts and the video for two consecutive probes, edibles were no longer provided. Video modeling and edibles were used for Henry for Zingo™ only and were introduced during the fifth intervention session. Vocal prompts were introduced for Joshua during the 20th session of Zingo™, the 54th intervention session of Memory™, and the 33th session of Fruit Ninja™. This occurred for Sean during session 20 of Memory™.

Generalization. The occurrence of scripted statements with a peer who was not a part of training sessions (i.e., a participant from another dyad) was assessed during generalization sessions. However, it is important to note that this peer learned a different portion of the script using the same activity. Untrained dyads were Leila and Joshua, Adam and Sean, and Terry and Henry. These sessions took place once every three to six sessions across baseline and intervention conditions and during both the 1-week and 3-week maintenance conditions.

Maintenance. The occurrence of scripted statements was evaluated 1 and 3 weeks after mastery. Participants did not have access to the intervention or activities between mastery and maintenance conditions.

Social validity. Twenty-seven teachers rated the social validity of procedures and outcomes. Teachers, who were familiar with the children, rated a written description of the procedures using the Treatment Acceptability Rating Form Revised (Reimers, Wacker, & Cooper, 1991). To assess the social validity of the outcomes, a video of Leila and Adam playing Memory™, pre- and post-intervention, was presented to teachers along with a survey assessing the quality and frequency of vocal statements. The survey included six questions: Four questions assessed responses using a 7-point Likert-type scale and two questions were open-ended where respondents could leave comments. One video was chosen from baseline and one video from maintenance. Videos were presented for only two of the six children to decrease the effort of responding and to prevent fatigue. Respondents were provided with an opportunity to respond to questions after viewing each video.

Treatment integrity. Treatment integrity data were collected by a teacher from the school in vivo for 50% of sessions across all phases of the study. The experimenter trained the teacher on data collection procedures using a checklist of defined and specified steps (e.g., presenting the iPad™). The experimenter modeled how to record data and provided opportunities for the teacher to practice data collection and receive feedback. Training was considered complete once the teacher independently scored two videos at 100%. Treatment integrity data were calculated by dividing the number of steps correctly implemented by the total number of steps and multiplying it by 100% to yield a percentage. Treatment integrity was 100% across participants and conditions. A secondary observer scored treatment integrity for 41% (Leila and Adam), 52% (Terry and Sean), and 54% (Henry and Joshua) of the sessions. Agreement between observers was also 100%.

RESULTS

Figures 1–3 show the effects of video modeling on emission of scripted statements for Leila and Adam, Terry and Sean, and Henry and Joshua, respectively. Across activities, none of the participants emitted any scripted statements during baseline. Following the introduction of video modeling, Leila, Adam, and Terry acquired the scripted statements with video modeling alone and demonstrated the skills when the video was faded. As shown in Figure 1, when video modeling was introduced, the percentage of scripted statements increased and were mastered within 2 sessions (Candyland™), 14 sessions (Memory™), and 6 sessions (Fruit Ninja™) for Leila and within 11 sessions (Candyland™), 20 sessions (Memory™), and 10 sessions (Fruit Ninja™) for Adam. Figure 2 shows the increase in the percentage of scripted statements for Terry following video modeling which resulted in mastery of activities across 11 sessions (Memory™), 8 sessions (Zingo™), and 9 sessions (Fruit Ninja™). Scripted statements increased from baseline following the introduction of video modeling in the presence of peers not associated with training for all participants. In addition, scripted statements maintained 1 and 3 weeks after mastery criterion across participants and games. For one participant (Sean), maintenance data fell to 70% during the initial maintenance assessment for one game (Memory™) but increased to 90% during the 3-week follow-up. As shown in Figures 2 and 3, the three remaining participants (i.e., Sean, Henry, and Josh) acquired scripted statements after procedural modifications were introduced. Sean required modeling and edible reinforcement across the first two activities (Memory™ and Zingo™). Edible reinforcement was introduced for Henry for the initial activity (Zingo™) whereas Joshua required edible reinforcement and modeling across all three of the activities.

On-task data were collected across dyads during baseline and experimental conditions (data are available upon request from the corresponding author). Across participants and

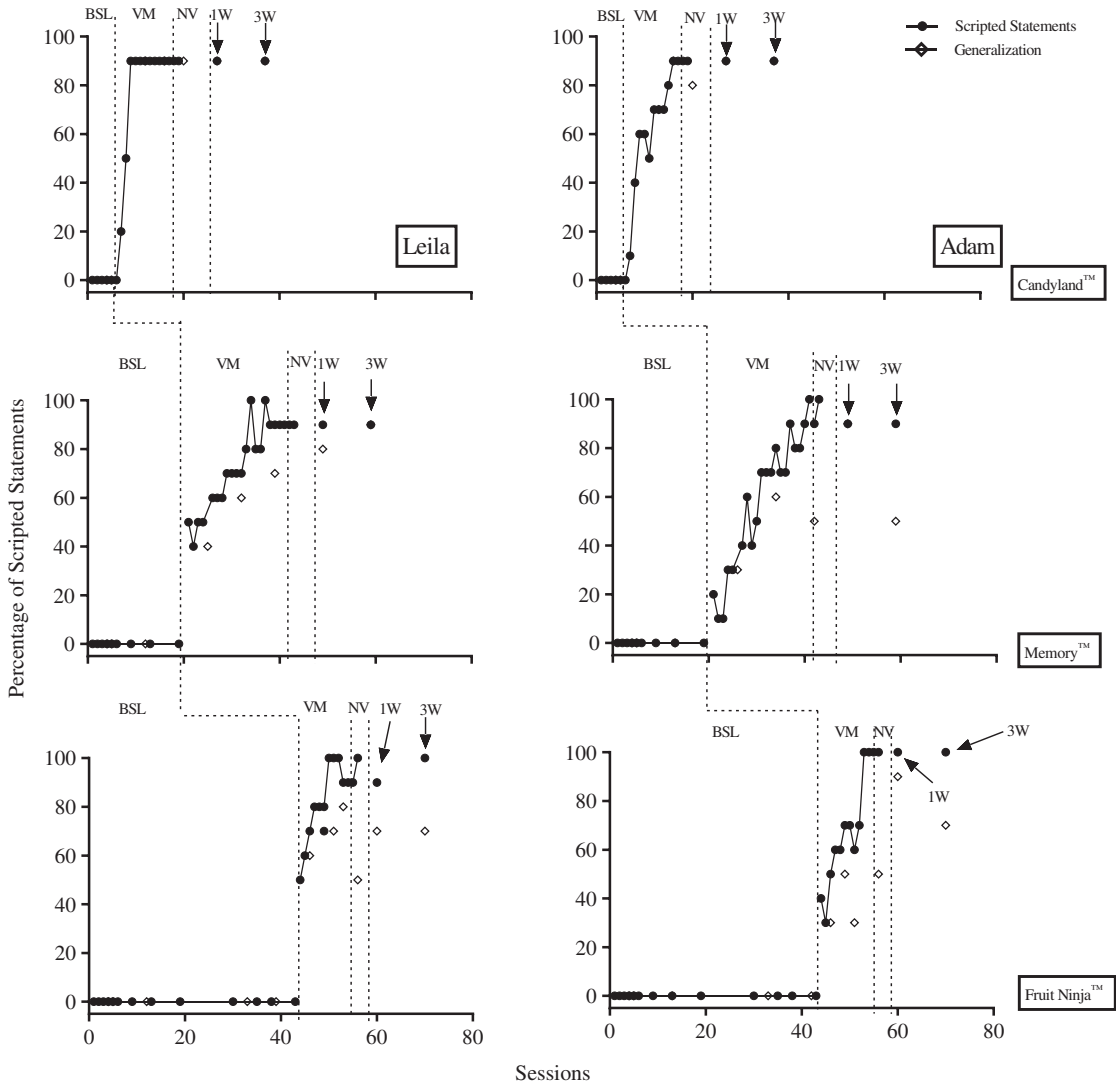


Figure 1. Scripted statements for Leila (left) and Adam (right) across Candyland™ (the first tier), Memory™ (the second tier), and for Fruit Ninja™ (the third tier). BSL = baseline phase, VM = the video modeling phase, and NV = the no video phase. 1 W = 1-week maintenance condition; 3 W = 3-week maintenance condition.

activities, on-task behavior during baseline was variable and ranged from 30% to 100% for Leila, from 15% to 100% for Adam, from 0% to 100% for Terry, and from 0% to 100% for Sean. On-task behavior during baseline for Henry was between 10% to 100% and between 0% to 100% for Joshua. On-task performance increased to 100% across participants

and games once video modeling was introduced.

Figure 4 shows the median social validity ratings of outcomes as well as the ratings provided by each teacher. As shown in Figure 4, the median ratings of Leila and Adam's baseline and postmastery conversation skills were 1 (i.e., the participant is not appropriately engaged)

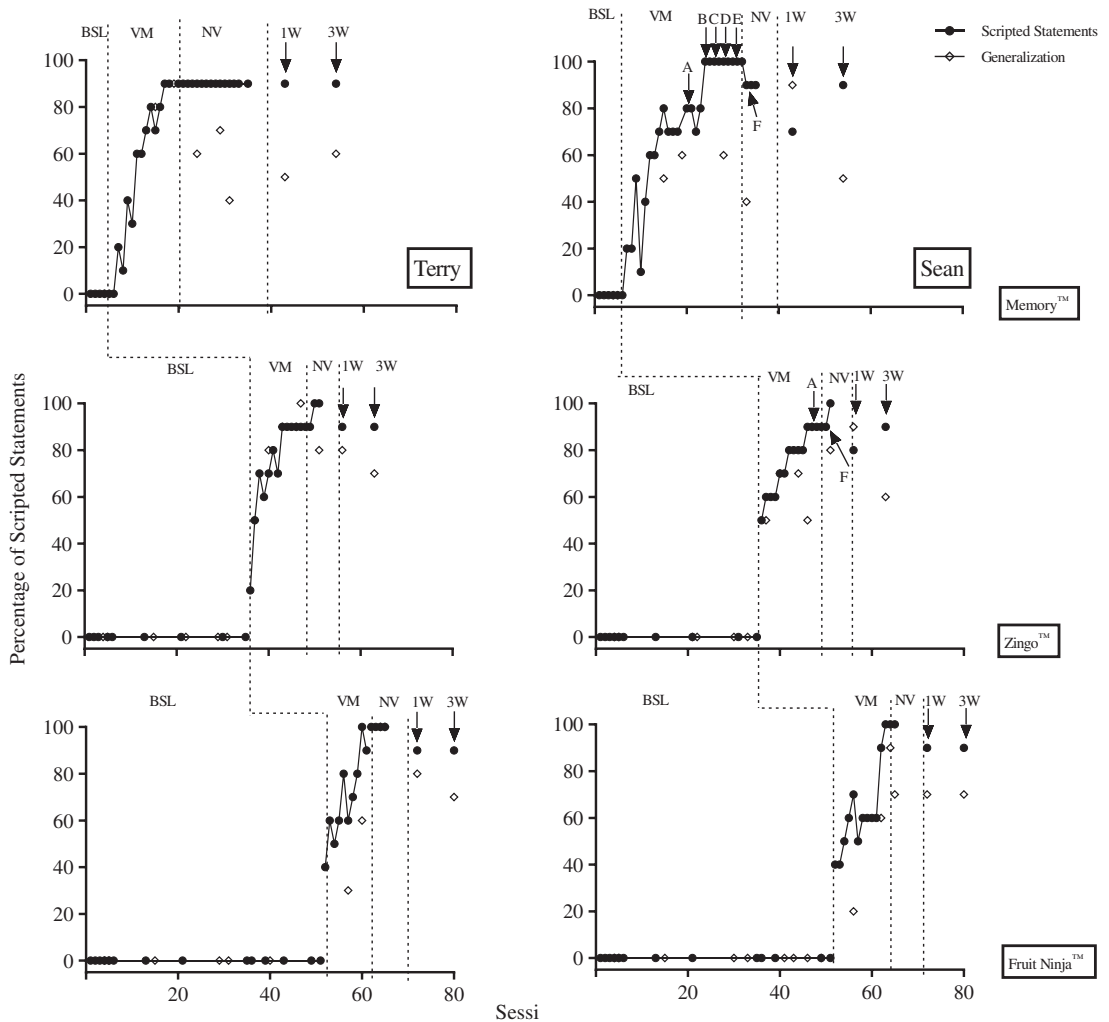


Figure 2. Scripted statements for Terry (left) and Sean (right) across Memory™ (the first tier), Zingo™ (the second tier), and Fruit Ninja™ (the third tier). BSL = the baseline phase, VM = the video modeling phase, and NV = the no video phase. 1 W = 1-week maintenance condition; 3 W = 3-week maintenance condition. A = the introduction of edibles, B = full prompts, C = partial prompts, D = phonemes, E = the removal of prompts, and F = the removal of edibles.

and 7 (i.e., the participant is appropriately engaged), respectively. Similarly, median ratings of Leila and Adam's play skills were 2 during baseline but improved to 7 (Leila) or 6 (Adam) during postmastery. Overall, teachers rated the procedures favorably (see Supporting Information 1). At least 89% of respondents scored ratings of 6 or higher on five questions pertaining to the acceptability and willingness to adopt the procedures. Additionally, at least 85% of

teachers rated three questions pertaining to effectiveness of procedures with a ranking of 6 or higher. Moreover, at least 89% of teachers rated the procedures to be cost effective with ranking of at most 2.

DISCUSSION

Our results aligned with previous research by showing that video modeling alone is effective

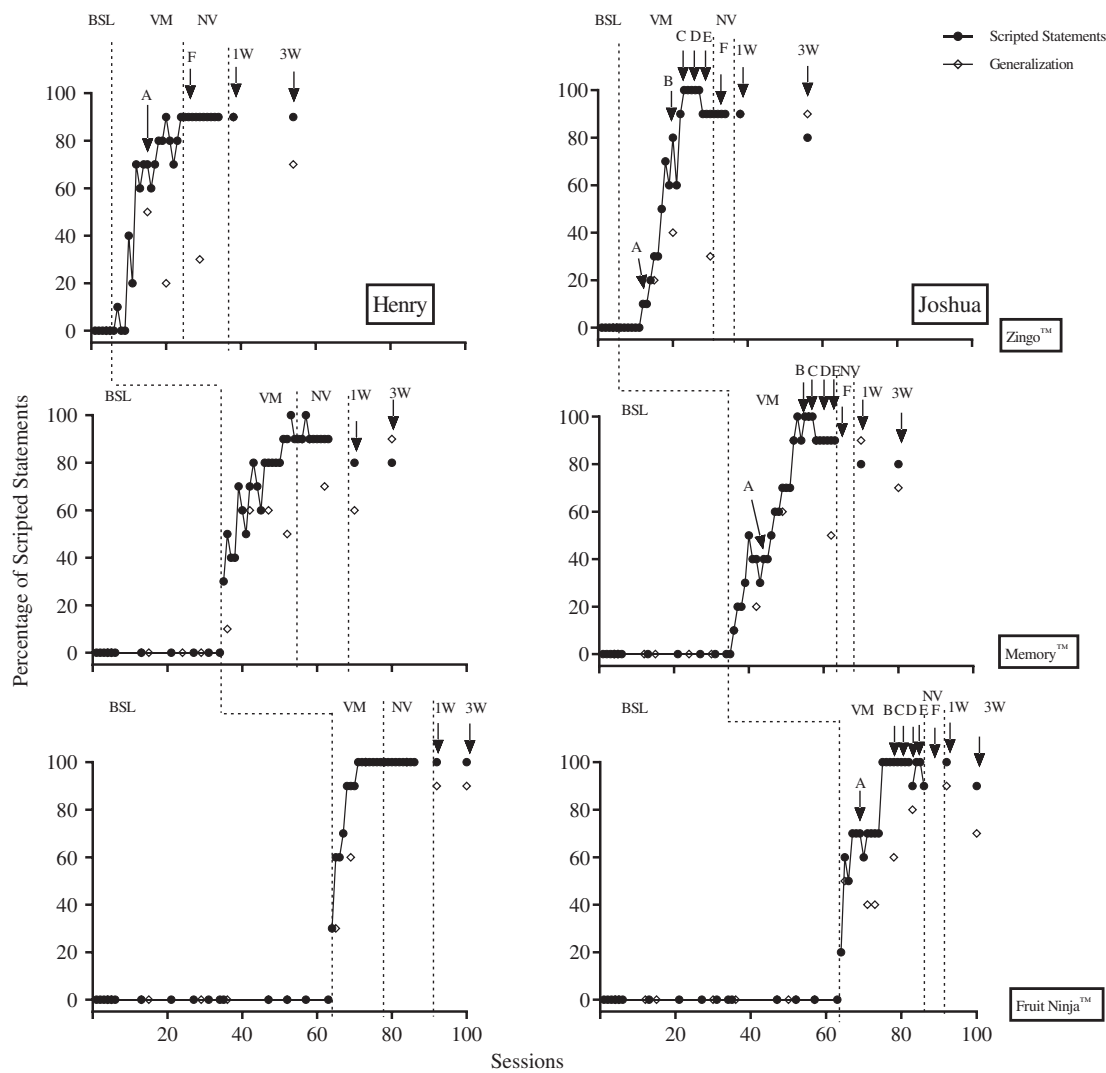
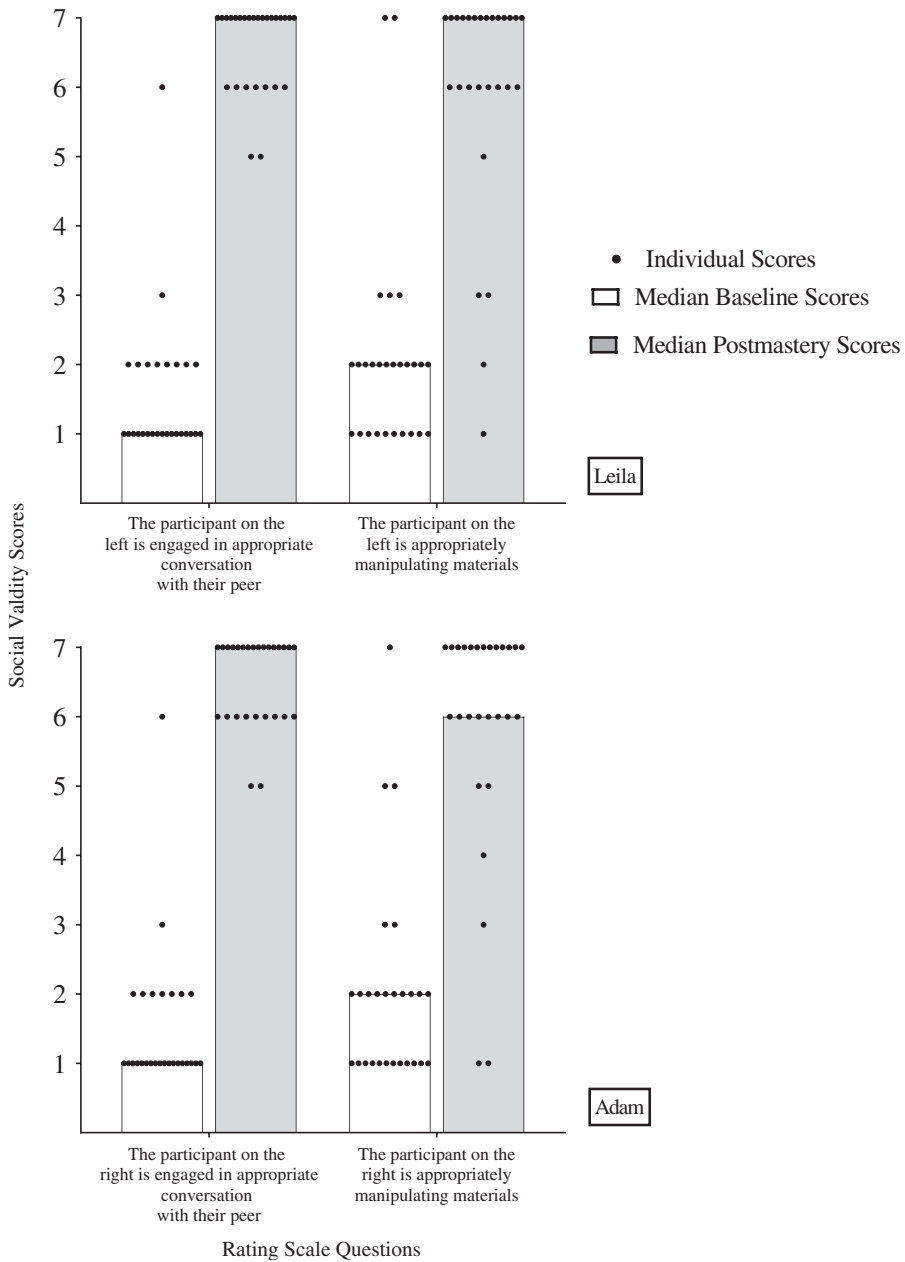


Figure 3. Scripted statements for Henry (left) and Joshua (right) across Zingo™ (the first tier), Memory™ (the second tier), and Fruit Ninja™ (the third tier). BSL = baseline phase, VM = the video modeling condition; NV = the no-video phase. 1 W = 1-week maintenance condition; 3 W = 3-week maintenance condition, A = the introduction of edibles, B = full prompts, C = partial prompts, D = phonemes, E = the removal of prompts, and F = the removal of edibles.

for some children with ASD (e.g., MacDonald et al., 2005; Nikopoulos & Keenan, 2007), although others may require additional prompts or reinforcers (e.g., Kourassanis et al., 2015; Nikopoulos & Keenan, 2003). Additionally, our generalization and maintenance outcomes align with previous research showing that skills taught through video modeling can generalize

and maintain (e.g., Nikopoulos & Keenan, 2003, 2007). Further, most teachers approved of the procedures and outcomes, which has also been found in previous video modeling research (e.g., Carlisle et al., 2018).

There are several variables that may have contributed to the success of the video modeling intervention. For one, responsiveness to



video-based instruction is affected by an individual's imitative repertoire. All six participants were reported to have a generalized imitative repertoire, which was verified with a

preassessment. Two of the six participants required preteaching despite having a history with video modeling. It remains important to identify variables contributing to the success of

video-based instruction. For example, individual learning histories with respect to imitating models may impact responsiveness to video-based instruction. Second, we included preferred, mastered activities. The extent to which preference for, and fluency with, the activity impacted our results remain unknown. Future research should more carefully consider the role of each when teaching communication skills, especially when teaching across multiple students.

There are several limitations of the present study. Procedural modifications were introduced based upon visual analysis of the data. Future research should consider stricter guidelines prior to introducing additional treatment components. In addition, the lack of an error correction procedure may have impacted the results and contributed to the need for procedural modifications. Additional research should be conducted to evaluate variations of error correction procedures when teaching behavior chains using video modeling. Although our social validity measures were robust, teachers completed the survey after viewing videos of sessions for one dyad (i.e., Leila and Adam). Although positive ratings were acquired, the social validity findings are limited to these participants. It is possible that respondents may not have rated the other participants' performance as favorable. In addition, contrived reinforcement (edibles) was needed for three of the participants (Henry, Sean, and Joshua). Although preference for social consequences was not directly assessed, it may be that contrived reinforcement was not needed for some participants (Leila, Adam, and Terry) because their behavior was sensitive to social consequences. More specifically, for some participants, social consequences may have functioned as reinforcers for scripted statements. Leila, Adam, and Terry were reported by classroom teachers to independently engage in social interactions across instructors and peers. For the other participants who required contrived

reinforcement, interactions with peers may not have functioned as a reinforcer. Additionally, following the completion of the study, Leila, Sean, and Henry approached the first author and requested to learn new games with their peers. This may suggest that social consequences have a role in the acquisition of scripted statements. Overall, future research should evaluate the role of sensitivity to social consequences when targeting social skills like commenting.

Another important future research direction is to identify assessments that predict the effectiveness of video modeling. Our preassessment did not accurately predict sensitivity to video modeling as procedural modifications were required for some participants. It is likely that our assessment was not predictive of successful teaching methods because targets used during our preassessment required fewer steps and may have been less complex than the behavior chains targeted during intervention. If the preassessment matched the complexity of the target skills, it may have better predicted sensitivity to video-based instruction. Researchers should continue evaluating prerequisite skills for video-based instruction. For example, delayed imitation with objects has been shown to be a strong predictor of the effectiveness of video modeling (MacDonald, Dickson, Martineau, & Ahearn, 2015). Although teachers reported that our participants engaged in delayed imitation, we did not directly assess it.

Performance with conversational partners not directly associated with training was variable across participants. The social skill deficits of the reciprocal partner may have impeded commenting during generalization probes. The use of a single script per activity may have contributed to rigidity in responding across participants. Perhaps instructional control of the conversational partner or the strict adherence to the script negatively impacted generalized outcomes with a social partner who learned a

different script. Future research might explore the effects of teaching multiple exemplars to maximize response and stimulus generalization. Assessing generalization with peers naïve to experimental procedures, and novel peers with a minimal level of social competence should also be considered. We also assessed whether participants made scripted statements when playing with a peer who was not part of training and whether they continued to make statements during play 1 and 3 weeks after mastery. Similar to previous research (e.g., Jones et al., 2014), trained responses did not always occur in untrained contexts. Future research should continue to evaluate the conditions under which trained skills occur with others who may not be a part of training but regularly interact with individuals with ASD (e.g., parents, siblings, or other family members).

Although video modeling led to an increase in scripted statements, no meaningful changes were observed in unscripted statements (data available upon request). Despite the lack of increase in unscripted statements, teaching scripted statements allowed a meaningful, low-effort response to contact socially mediated consequences. Over time and paired with other primary reinforcers and responses (Dozier, Iwata, Thomason-Sassi, Worsdell, & Wilson, 2012), these social interactions may become conditioned reinforcers. However, we did not assess this in our current study. Poor outcomes for unscripted statements are not surprising because we did not program for generalization or novel recombination of scripted statements; however, we taught scripted comments across three activities. Future research should evaluate the effects of teaching multiple variations of contextually relevant statements across the same activities to enhance response generalization across activities. Specifically, this could be accomplished by teaching variations of scripted statements such as “This game is awesome,” “I love this game!” or “This game is the best!” Researchers should explore strategies that could

promote novel recombination of scripts or strategies that teach multiple contextually appropriate scripted statements (i.e., matrix training). Although data during generalization probes were variable, teachers reported that participants engaged in scripted statements during novel games in the classroom. It is likely that the generalization assessment failed to capture appropriate motivating and discriminative functions of environmental variables that evoke social reciprocity. Future research should explore the variables that may impact motivation to socially comment and meaningfully engage in leisure activities.

We developed our scripts by observing peers of typical development. Previous research lacked details regarding the development of scripts (e.g., MacDonald et al., 2005; McGee et al., 1984; Nikopoulos & Keenan, 2003; & Reagon et al., 2006). There may be collateral benefits to using age- and contextually appropriate statements. That is, peer-derived targets (i.e., scripted statements) may be more socially valid than arbitrarily selected ones and more likely to contact reinforcement in the typical environment. Researchers should consider using age-matched peers of typical development as video models (e.g., Kourassanis et al., 2015). We did attempt to use age-matched peers; however, our models required substantial support to state the script and to remain on task during filming. Due to these challenges, age-matched peers of typical development were not included.

Although previous research has shown the benefits of using video modeling to teach children with ASD (e.g., MacDonald et al., 2005), comparisons of video modeling and other instructional strategies like behavior skills training (e.g., Nuernberger, Ringdahl, Vargo, Crumpecker, & Gunnarsson, 2013), script and script fading (e.g., Krantz & McClannahan, 1993), joint activity schedules (e.g., Betz et al., 2008), in-vivo modeling (e.g., McGee et al., 1984) should be conducted. Since we

incorporated previously mastered activities, the effects of video modeling on scripted statements with novel activities remain unknown and should be considered in future explorations. Procedures may also be replicated with individuals who are older, across different types of leisure activities, or with those who use augmentative or alternative communication.

Video modeling remains an effective instructional strategy across several domains important for independence and enhancing quality of life, like conversational skills. Our results are promising in that we established meaningful repertoires across multiple learners using video modeling. However, there is more to learn about generalization to untrained peers.

REFERENCES

- American Psychiatric Association (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Retrieved from <http://dsm.psychiatryonline.org/book.aspx?bookid=556>
- Apple, A. L., Billingsley, F., & Schwartz, I. S. (2005). Effects of video modeling alone and with self-management on compliment-giving behaviors of children with high-functioning ASD. *Journal of Positive Behavior Interventions*, 7, 33–46. doi.org/10.1177%2F10983007050070010401
- Betz, A., Higbee, T. S., & Reagon, K. A. (2008). Using joint activity schedules to promote peer engagement in preschoolers with autism. *Journal of Applied Behavior Analysis*, 41, 237–241. <https://doi.org/10.1901/jaba.2008.41-237>.
- Carlile, K. A., DeBar, R. M., Reeve, S. A., Reeve, K. F., & Meyer, L. S. (2018). Teaching help-seeking when lost to individuals with autism spectrum disorder. *Journal of Applied Behavior Analysis*, 51, 191–206. <https://doi.org/10.1002/jaba.447>.
- Carr, J. E., Nicolson, A. C., & Higbee, T. S. (2000). Evaluation of a brief multiple-stimulus preference assessment in a naturalistic context. *Journal of Applied Behavior Analysis*, 33, 353–357. <https://doi.org/10.1901/jaba.2000.33-353>.
- Charlop, M. H., & Milstein, J. P. (1989). Teaching autistic children conversational speech using video modeling. *Journal of Applied Behavior Analysis*, 22, 275–285. <https://doi.org/10.1901/jaba.1989.22-275>.
- Dozier, C. L., Iwata, B. A., Thomason-Sassi, J., Worsdell, A. S., & Wilson, D. M. (2012). A comparison of two pairing procedures to establish praise as a reinforcer. *Journal of Applied Behavior Analysis*, 45, 721–735. <https://doi.org/10.1901/jaba.2012.45-721>.
- Dunn, L. M., & Dunn, D. M. (2007). *Peabody picture vocabulary test: PPVT-4A*. Minneapolis, MN: NCS Pearson.
- Fisher, W. W., Piazza, C. C., Bowman, L. G., & Amari, A. (1996). Integrating caregiver report with a systematic choice assessment. *American Journal on Mental Retardation*, 101, 15–25. Retrieved from <https://www.ncbi.nlm.nih.gov/pubmed/8827248>
- Gilliam, J. E. (2013). *Gilliam autism rating scale* (3rd ed.). Austin, TX: Pro-Ed. Retrieved from pearsonclinical.com.
- Green, V. A., Drysdale, H., Boelema, T., Smart, E., van der Meer, L., Achmadi, D., ... Lancioni, G. (2013). Use of video modeling to increase positive peer interactions of four preschool children with social skills difficulties. *Education and Treatment of Children*, 36, 59–85. <https://doi.org/10.1353/etc.2013.0016>.
- Hagopian, L. P., Rush, K. S., Lewin, A. B., & Long, E. S. (2001). Evaluating the predictive validity of a single stimulus engagement preference assessment. *Journal of Applied Behavior Analysis*, 34, 475–485. <https://doi.org/10.1901/jaba.2001.34-475>.
- Horner, R. D., & Baer, D. M. (1978). Multiple-probe technique: A variation on the multiple baseline. *Journal of Applied Behavior Analysis*, 11, 189–196. <https://doi.org/10.1901/jaba.1978.11-189>.
- Jones, J., Lerman, D. C., & Lechago, S. (2014). Assessing stimulus control and promoting generalization via video modeling when teaching social responses to children with autism. *Journal of Applied Behavior Analysis*, 47, 37–50. <https://doi.org/10.1002/jaba.81>.
- Kourassanis, J., Jones, E. A., & Fienup, D. M. (2015). Peer-video modeling: Teaching chained social game behaviors to children with ASD. *Journal of Developmental and Physical Disabilities*, 27, 25–36. <https://doi.org/10.1007/s10882-014-9399-8>.
- Krantz, P. J., & McClannahan, L. E. (1993). Teaching children with autism to initiate to peers: Effects of a script-fading procedure. *Journal of Applied Behavior Analysis*, 26, 121–132. <https://doi.org/10.1901/jaba.1993.26-121>.
- LeBlanc, L. A., Coates, A. M., Daneshvar, S., Charlop-Christy, M. H., Morris, C., & Lancaster, B. M. (2003). Using video modeling and reinforcement to teach perspective-taking skills to children with autism. *Journal of Applied Behavior Analysis*, 36, 253–257. <https://doi.org/10.1901/jaba.2003.36-253>.
- MacDonald, R., Clark, M., Garrigan, E., & Vangala, M. (2005). Using video modeling to teach pretend play to children with autism. *Behavioral Interventions*, 20, 225–238. <https://doi.org/10.1002/bin.197>.
- MacDonald, R., Sacramone, S., Mansfield, R., Wiltz, K., & Ahearn, W. H. (2009). Using video modeling to teach reciprocal pretend play to children with autism. *Journal of Applied Behavior Analysis*, 42, 43–55. <https://doi.org/10.1901/jaba.2009.42-43>.

- MacDonald, R. P. F., Dickson, C. A., Martineau, M., & Ahearn, W. H. (2015). Prerequisite skills that support learning through video modeling. *Education and Treatment of Children, 38*, 3348. <https://doi.org/10.1353/etc.2015.0004>.
- McGee, G. G., Almeida, M. C., Sulzer-Azaroff, B., & Feldman, R. S. (1992). Promoting reciprocal interactions via peer incidental teaching. *Journal of Applied Behavior Analysis, 25*, 117–126. <https://doi.org/10.1901/jaba.1992.25-117>.
- McGee, G. G., Krantz, P. J., & McClannahan, L. F. (1984). Conversational skills for autistic adolescents: Teaching assertiveness in natural game settings. *Journal of Autism and Developmental Disorders, 14*, 319–330. <https://doi.org/10.1007/BF02409582>.
- Murdock, L. C., Ganz, J., & Crittendon, J. (2013). Use of an iPad play story to increase play dialogue of preschoolers with autism spectrum disorder. *Journal of Autism and Developmental Disorders, 43*, 2174–2189. <https://doi.org/10.1007/s10803-013-1770-6>.
- Nikopoulos, C. K., & Keenan, M. (2003). Promoting social initiation in children with autism using video modeling. *Behavioral Interventions, 18*, 87–108. <https://doi.org/10.1002/bin.129>.
- Nikopoulos, C. K., & Keenan, M. (2007). Using video modeling to teach complex social sequences to children with autism. *Journal of Autism and Developmental Disorders, 37*, 678–693. <https://doi.org/10.1007/s10803-006-0195-x>.
- Nuernberger, J. E., Ringdahl, J. E., Vargo, K. K., Crumpecker, A. C., & Gunnarsson, K. F. (2013). Using a behavioral skills training package to teach conversation skills to young adults with autism spectrum disorders. *Research in Autism Spectrum Disorders, 7*, 411–417. <https://doi.org/10.1016/j.rasd.2012.09.004>.
- Reagon, K. A., Higbee, T. S., & Endicott, K. (2006). Teaching pretend play skills to a student with autism using video modeling with a sibling as a model and play partner. *Education and Treatment of Children, 29*, 517–528. Retrieved from https://www.jstor.org/stable/42899899?seq=1#page_scan_tab_contents
- Reimers, T. M., Wacker, D. P., & Cooper, L. J. (1991). Evaluation of the acceptability of treatments for children's behavioral difficulties: Ratings by parents receiving services in an outpatient clinic. *Child & Family Behavior Therapy, 13*, 53–71. https://doi.org/10.1300/J019v13n02_04.
- Shukla-Mehta, S., Miller, T., & Callahan, K. J. (2010). Evaluating the effectiveness of video instruction on social and communication skills training for children with autism spectrum disorders: A review of the literature. *Focus on Autism and Other Developmental Disabilities, 25*, 23–36. <https://doi.org/10.1177/1088357609352901>.
- Sundberg, M. L. (2008). *Verbal behavior milestones assessment and placement program: The VB-MAPP*. Concord, CA: AVB Press.
- Taylor, B. A., Levin, L., & Jasper, S. (1999). Increasing play-related statements in children with autism towards their siblings: Effects of video modeling. *Journal of Developmental and Physical Disabilities, 11*, 253–264. <https://doi.org/10.1023/A:1021800716392>.
- Torres, R., DeBar, R. M., Reeve, S. A., Meyer, L. S., & Covington, T. M. (2018). The effects of a video-enhanced schedule on exercise behavior. *Behavior Analysis in Practice, 11*, 85–96. <https://doi.org/10.1007/s40617-018-0224-1>.
- Williams, K. T. (2007). *Expressive vocabulary test* (2nd ed.). Circle Pines, MN: AGS Publishing.

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